The drone is powered by a single Rolls-Royce Turbomeca Adour engine, providing subsonic performance suitable for stealth operations. Internal air intakes and exhaust ducts are carefully shaped to reduce radar and infrared signatures. The internal weapons bay allows the Neuron to carry precision-guided munitions (PGMs), including 250 kg-class smart bombs, while maintaining its stealth profile (De Saint-Jean, et al. 2013).

A key design feature of the Neuron is its fully autonomous flight management system, capable of executing mission profiles without continuous operator control. The system integrates data fusion, mission reconfiguration, and real-time decision-making algorithms, allowing the UCAV to adapt to dynamic combat environments.

Technological features and innovations

1. *Stealth technology*: The Neuron employs radar-absorbent materials (RAM), internal weapon carriage, and smooth aerodynamic surfaces to achieve a low radar cross-section suitable for deep penetration missions.
2. *Autonomous operations*: Advanced mission computers enable autonomous navigation, target identification, and threat avoidance, reducing operator dependency and improving response time.
3. *Modular mission architecture*: The Neuron's open avionics system supports multiple mission configurations, from strike operations to reconnaissance, by integrating different sensor and weapon payloads.
4. *Fly-by-light control system*: A notable innovation is the fly-by-light (FBL) system, an optical data transmission network offering high resistance to electromagnetic interference and improved reliability compared to traditional fly-by-wire systems.
5. *Collaborative development framework*: The project unified European aerospace expertise, with Dassault leading design, Saab managing